Proof of the right distributive law is a similar computation. Thus S is a ring.

b. We have $(0,1)(r,n) = (0r + 1 \cdot r + n \cdot 0, 1n) = (r,n) = (r0 + n \cdot 0 + 1 \cdot r, n1) = (r,n)(0,1)$, so $(0,1) \in S$ is unity.

c. Using Theorem 19.15 and part **b**, the ring S either has characteristic 0 or the smallest positive integer n such that $(0,0) = n \cdot (0,1) = (0,n)$. Clearly n has this property if and only if $S = R \times \mathbb{Z}_n$. Because we chose $\mathbb{Z}$ or $\mathbb{Z}_n$ to form S acccording as R has characteristic 0 or n, we see that R and S have the same characteristic.

d. We have $\phi(r_1 + r_2) = (r_1 + r_2, 0) = (r_1, 0) + (r_2, 0) = \phi(r_1) + \phi(r_2)$. Also, $\phi(r_1 r_2) = (r_1 r_2, 0) = (r_1 r_2 + 0 \cdot r_2 + 0 \cdot r_1, 00) = (r_1, 0)(r_2, 0) = \phi(r_1)\phi(r_2)$. Thus ϕ is a homomorphism. If $\phi(r_1) = \phi(r_2)$, then $(r_1, 0) = (r_2, 0)$ so $r_1 = r_2$; Thus ϕ is one to one. Therefore ϕ maps R isomorphically onto the subring $\phi[R]$ of S.

20. Fermat's and Euler's Theorems

1. Either 3 or 5 **2.** Either 2, 6, 7, or 8 **3.** Either 3, 5, 6, 7, 10, 11, 12, or 14

4. $3^{47} \equiv (3^{22})^2 \cdot 3^3 \equiv 1^2 \cdot 27 \equiv 27 \equiv 4 \pmod{23}$ **5.** $37^{49} \equiv 2^{49} \equiv (2^6)^8 \cdot 2 \equiv 1^8 \cdot 2 \equiv 2 \pmod{7}$

6. $2^{17} \equiv (2^4)^4 \cdot 2 \equiv (-2)^4 \cdot 2 \equiv 16 \cdot 2 \equiv 14 \pmod{18}$. Thus $2^{17} = 18q + 14$. Then $2^{(2^{17})} \equiv 2^{18q+14} \equiv (2^{18})^q \cdot 2^{14} \equiv 1^q \cdot 2^{14} \equiv (2^4)^3 \cdot 2^2 \equiv (-3)^3 \cdot 2^2 \equiv -27 \cdot 4 \equiv -8 \cdot 4 \equiv 6 \pmod{19}$ so the answer is $6 + 1 = 7$.

7. (See the answer in the text.)

8. All positive integers less than p^2 that are not divisible by p are relatively prime to p. Thus we delete from the $p^2 - 1$ integers less than p^2 the integers $p, 2p, 3p, \cdots, (p-1)p$. There are $p-1$ integers deleted, so $\phi(p^2) = (p^2 - 1) - (p - 1) = p^2 - p$.

9. We delete from the $pq - 1$ integers less than pq those that are mltiples of p or of q to obtain those relatively prime to pq. The multiples of p are $p, 2p, 3p, \cdots, (q-1)p$ and the multiples of q are $q, 2q, 3q, \cdots, (p-1)q$. Thus we delete a total of $(q-1) + (p-1) = p + q - 2$ elements, so $\phi(pq) = (pq - 1) - (p + q + 2) = pq - p - q + 1 = (p-1)(q-1)$.

10. From Exercise 7, we find that $\phi(24) = 8$, so $7^8 \equiv 1 \pmod{24}$. Then $7^{1000} \equiv (7^8)^{125} \equiv 1^{125} \equiv 1 \pmod{24}$.

11. We can reduce the congruence to $2x \equiv 2 \pmod{4}$. The gcd of 4 and 2 is $d = 2$ which divides $b = 2$. We divide by 2 and solve instead the congruence $x \equiv 1 \pmod{2}$. Of course $x = 1$ is a solution. Another incongruent (mod 4) solution is $x = 1 + 2 = 3$. Thus the solutions are the numbers in $1 + 4\mathbb{Z}$ and $3 + 4\mathbb{Z}$.

12. We can reduce the congruence to $7x \equiv 5 \pmod{15}$. The gcd of 15 and 7 is $d = 1$ which divides $b = 5$. By inspection, $x = 5$ is a solution, and all solutions must be congruent to 5. Thus the solutions are the numbers in $5 + 15\mathbb{Z}$.

13. The congruence can be reduced to $12x \equiv 15 \pmod{24}$. The gcd of 24 and 12 is $d = 12$ which does not divide $b = 15$, so there are no solutions.

14. The congruence can be reduced to $21x \equiv 15 \pmod{24}$. The gcd of 24 and 21 is $d = 3$ which divides $b = 15$. We divide by 3 and solve instead the congruence $7x \equiv 5 \pmod{8}$. By inspection, $x = 3$ is a solution. Other incongruent (mod 24) solutions are given by $x = 3 + 8 = 11$ and $x = 3 + 2 \cdot 8 = 19$. Thus the solutions are the numbers in $3 + 24\mathbb{Z}$, $11 + 24\mathbb{Z}$, or $19 + 24\mathbb{Z}$.

15. The congruence can be reduced to $3x \equiv 8 \pmod{9}$. The gcd of 9 and 3 is $d = 3$ which does not divide $b = 8$, so there are no solutions.

16. The congruence can be reduced to $5x \equiv 8 \pmod{9}$. The gcd of 9 and 5 is 1 which divides 8. By inspection, $x = 7$ is a solution, and there are no other incongruent (mod 9) solutions, so solutions are the numbers in $7 + 9\mathbb{Z}$.

17. The congruence can be reduced to $25x \equiv 10 \pmod{65}$. The gcd of 65 and 25 is $d = 5$ which divides $b = 10$. We divide by 5 and solve instead the congruence $5x \equiv 2 \pmod{13}$. By inspection $x = 3$ is one solution. The other solutions that are incongruent (mod 65) are $3 + 13 = 16$, $3+2\cdot 13 = 29$, $3+3\cdot 13 = 42$, and $3 + 4 \cdot 13 = 55$. Thus the solutions are the numbers in $3 + 65\mathbb{Z}, 16 + 65\mathbb{Z}, 29 + 65\mathbb{Z}, 42 + 65\mathbb{Z}$, and $55 + 65\mathbb{Z}$.

18. The gcd of 130 and 39 is $d = 13$ which divides $b = 52$. We divide by 13 and solve instead $3x \equiv 4 \pmod{10}$. By inspection, $x = 8$ is a solution. Repeatedly adding 10 eleven times, we see that the solutions are the numbers in $8 + 130\mathbb{Z}, 18 + 130\mathbb{Z}, 28 + 130\mathbb{Z}, 38 + 130\mathbb{Z}, 48 + 130\mathbb{Z}, 58 + 130\mathbb{Z}, 68 + 130\mathbb{Z}, 78 + 130\mathbb{Z}, 88 + 130\mathbb{Z}, 98 + 130\mathbb{Z}, 108 + 130\mathbb{Z}, 118 + 130\mathbb{Z}$, or $128 + 130\mathbb{Z}$.

19. Because $(p-1)! = (p-1)(p-2)!$, Exercise 28 shows that we have $-1 \equiv (p-1) \cdot (p-2)! \pmod{p}$. Reducing (mod p), we have the congruence $-1 \equiv (-1) \cdot (p-2)! \pmod{p}$, so we must have $(p-2)! \equiv 1 \pmod{p}$.

20. Taking $p = 37$ and using Exercise 28, we have

$$36! \equiv (36)(35)(34!) \equiv -1 \pmod{37}$$

so $(-1)(-2)(34!) \equiv -1 \pmod{37}$ and $2(34!) \equiv 36 \pmod{37}$. Thus $34! \equiv 18 \pmod{37}$.

21. Taking $p = 53$ and using Exercise 28, we have

$$52! \equiv (52)(51)(50)(49!) \equiv -1 \pmod{53}$$

so $(-1)(-2)(-3)(49!) \equiv -1 \pmod{53}$ and $6(49!) \equiv 1 \pmod{53}$. By inspection, we see that $49! \equiv 9 \pmod{53}$.

22. Taking $p = 29$ and using Exercise 28, we have

$$28! \equiv (28)(27)(26)(25)(24!) \equiv -1 \pmod{29}$$

so $(-1)(-2)(-3)(-4)(24!) \equiv -1 \pmod{29}$ and $(-5)(24!) \equiv -1 \pmod{29}$. By inspection, we see that $24! \equiv 6 \pmod{29}$.

23. F T T F T T F T F T

24.

$\cdot_{12}$	1	5	7	11
1	1	5	7	11
5	5	1	11	7
7	7	11	1	5
11	11	7	5	1

This group is isomorphic to $\langle \mathbb{Z}_2 \times \mathbb{Z}_2, + \rangle$.